# MERIDIAN ♥

## Memory-Encoded Rituals for Intimacy across Distance in an Interactive, Asynchronous Neighbourhood

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(a) The whole town, drawn in code (b) The square: guide, partner, cats (c) Prologue: the birthday cake (d) Finale: the wedding

**Figure 1: Yi Town at a glance. (a) The town map: a $24 \times 20$ grid of 16-pixel tiles with nine numbered houses, one per level, painted entirely by code. (b) The fountain square in play, top after every level is cleared (gold flags, a cat sunbathing by the pond) and bottom halfway through; the player meets the partner, the guide Rayford and the cats. (c) The birthday prologue: nine candles in the dark at Shenzhen midnight, and ten strawberries set in a heart. (d) The wedding in the fountain garden, officiated by Rayford with the two cats as guests.**

## Abstract

How can shared memories become playable structures for partners who live in different time zones? We present MERIDIAN, a pre-deployment design case of Yi Town, a bilingual browser game developed by one member of a long-distance couple as a gift for the other. Nine levels translate personally meaningful events, places and cultural references into tasks within a navigable pixel town. A birthday prologue and a fictional wedding frame the intended experience. Drawing on research through design, we reconstruct four design changes from 109 development commits, 28 changelog versions and three design specifications. These records support six situated design commitments, including memory-derived mechanics, staged access, local progress storage and bilingual coverage. The implementation combines a union-based progress model with manually exchanged, checksummed codes and time-zone-dependent scenery. An implementation audit reproduced 153 passing unit tests and examined all 1,024 canonical progress codes, while identifying limits involving resets, unversioned level registries and residual transcription errors. The application has no author-operated backend for progress, but external media services and browser speech recognition limit its privacy and offline guarantees. No recipient evaluation or relationship outcomes are reported. The contribution is an inspectable account of design decisions and their trade-offs, including the tension between authoring a personal gift and preserving the recipient's freedom to interpret, decline or reshape it. The game is playable at https://jingyifan.love and its source is public at https://github.com/rayford295/yi-town.

## Keywords

intimate computing, long-distance relationships, autobiographical design, research through design, games for two, local-first software, bilingual interfaces, pixel art, ceremony, surprise

## 1 Introduction

Long-distance partners use communication technologies to maintain awareness, exchange affection and share ordinary activities [20, 25, 33]. Some report substantial intimacy despite geographical separation, but such findings do not imply that a particular technology improves a relationship [15, 31]. This paper asks a narrower design question: how can a couple's shared material become an asynchronous game, and what tensions arise when one partner authors both its narrative and its interaction constraints?

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Yi Town is a browser game developed by Yifan for Jingyi and himself. They began their relationship on 20 September 2026 and live in Shenzhen, China, and College Station, Texas. Their clocks differed by thirteen hours during development; the difference becomes fourteen hours when Texas observes standard time. The game draws on shared songs, a Gomoku game, dated memories and a symbolic hand-drawn ticket representing a planned meeting. Players navigate a pixel town whose nine houses contain levels, while a birthday prologue, a letter and a fictional wedding organise the narrative (Figure 1). The game is built for separate visits; playing at the same time is possible but never required.

We use Meridian as the name for this design case: Memory-Encoded Rituals for Intimacy across Distance in an Interactive, Asynchronous Neighbourhood. A *memory-encoded ritual* means a scripted action whose intended significance derives from shared autobiographical material, and *intimacy* names the intended qualities of recognition, affection and shared meaning. The central design move is to make a memory shape an action or a rule, rather than only illustrate an interface.

The paper reports an artefact and a retrospective reconstruction of its development before the recipient's planned first play on 31 October 2026. It draws on research through design and autobiographical design, without claiming the extensive lived use associated with mature first-person studies [7, 26, 37]. Its contributions are:

(1) A documented design case that translates autobiographical material into nine game levels and staged ceremonies (Sections 3 and 4).
(2) Six situated design commitments and four traceable development episodes, with explicit distinctions between recorded changes, interpretations and unevaluated intentions (Table 2; Section 6).
(3) An implementation account and verification audit of local progress, manual code exchange and selected interface properties, together with their privacy and reliability limits (Section 5).

## 2 Related Work

### 2.1 Mediating Intimacy at a Distance

Vetere et al. studied how couples and families express intimacy and argued that technologies for strong ties should support the expressive, the physical and the playful, not only the informational [33]. Hassenzahl et al. reviewed 143 intimacy-mediating artefacts and grouped them into six strategies: awareness, expressivity, physicalness, gift giving, joint action and memories [11]. Many such systems are deliberately minimal channels, such as the Virtual Intimate Object, a single button whose meaning the couple negotiates [17]. Neustaedter and Greenberg found that some long-distance couples leave a video call running for hours to share "being together" rather than to talk [25]; CoupleVIBE conveyed a partner's location by vibration [3]; SyncDecor linked everyday appliances in two homes [32]. Yi Town sits in Hassenzahl's *memories* and *gift giving* strategies. What distinguishes it is what the interface represents: a collection of shared references organised as tasks rather than a live awareness signal.

### 2.2 Autobiographical and First-Person Design

Neustaedter and Sengers defined autobiographical design as design research drawing on genuine use by the people who built the system [26], and Desjardins and Ball documented its tensions, including the blurred line between designer and participant, and the intimacy of the data involved [5]. First-person methods include situated inquiry through the researcher's own experience; a small number of people alone does not establish methodological depth [6, 21]. Research through design treats the artefact itself as a carrier of knowledge [7, 37]. Yi Town uses autobiographical material and developer self-testing, but the present account precedes sustained use by the recipient. We therefore distinguish the making of the artefact from an evaluation of living with it.

### 2.3 Play, Ritual and Games for Two

Huizinga described play as a "magic circle" set apart from ordinary life [12]; Salen and Zimmerman made the circle a design tool [28]; Juul argued that games are half real rules and half fiction [16]. Sicart argued that play appropriates the world and its objects [30], and Isbister argued that games build connection between players through shared action [14]. Ludic design in HCI values curiosity and ambiguity over task efficiency [8, 9], and slow technology designs for reflection rather than throughput [10, 27]. Aron et al.'s procedure for generating closeness through escalating mutual disclosure [2] is, in a sense, a level structure. Yi Town takes the game form literally: locks, levels, a guide, a prologue and a finale. Bespoke ritual technologies already explore connection across distance [18]; our case organises selected memories and imagined futures as a sequence of game actions.

### 2.4 Local-First and Backend-Free Progress

Local-first software seeks to combine collaboration with user control and durable access to data [19]. State-based conflict-free replicated data types (CRDTs) converge when replicas combine compatible state through an appropriate merge; the grow-only set uses union [29]. Yi Town adopts this operation for level completion and exchanges state manually. Progress is local, but static hosting, remote media and browser-dependent recognition remain part of its infrastructure. Keeping progress local also does not protect personal images that are distributed as static assets; visual content can expose location information [35].

## 3 Design Context and Method

### 3.1 The Two of Us

Table 1 lists the dates that the system encodes. The couple speak Mandarin to each other; Yifan works in English, Jingyi in Chinese and English. Both use phones as their main devices, and Jingyi's network in mainland China blocks some services (YouTube, Google Fonts) that Yifan's does not. These circumstances motivated bilingual text, alternative media providers and a phone-oriented interface.

### 3.2 Scope, Corpus and Reconstruction

The design corpus comprises 109 commits dated 24–28 October 2026, 28 changelog entries from version 0.1.0 to 1.17.0, and three

**Table 1: Key dates encoded in the system (`src/core/config.ts`).**

| Date | Moment |
|---|---|
| 2026-09-20 | We met; day 1 of every counter in the town |
| 2026-09-25 | Yifan's birthday (Libra) |
| 2026-10-17 | First video call; the hand-drawn ticket CYF925 |
| 2026-10-23 | First Gomoku game, on a Zoom whiteboard |
| 2026-10-24 | First commit of Yi Town |
| 2026-10-31 | Jingyi's birthday (Scorpio); intended first recipient play |
| 2027-02-05 | Planned meeting; symbolic ticket CYF925; countdown target |

specifications for the site, quest and town. The development cut-off is commit `55440ae`. Yifan directed the design, supplied the autobiographical material and tested both player roles, with substantial implementation assistance from an AI coding assistant. Jingyi is the intended recipient. This account reports no interview, diary, recipient play session or relationship measure.

For the present reconstruction, we inspect the documented changes underlying four episodes: replacing Pictionary, changing the ring's completion condition, correcting represented routines, and revising the entry gate (Section 6). Each episode links a source change to an interpretation of the design trade-off. The selection illustrates decisions discussed in this paper rather than an exhaustive coding of all commits. Version history establishes what changed and when, not the designer's unrecorded motivation or the recipient's response.

## 3.3 Knowledge Claims and Positionality

Research through design can make an artefact and its rationale available for scrutiny without claiming a universal solution [7, 37]. First-person inquiry also requires attention to whose experience becomes evidence [5, 6]. Here, the designer's familiarity with the relationship informs the content but cannot stand in for the other partner's perspective. The six commitments below are situated design choices reconstructed from the artefact and log, not validated principles.

## 3.4 Six Design Commitments

We articulate six commitments from this reconstruction of the five-day development period. Table 2 maps each to the mechanisms that realise it.

D1 Memory-derived mechanics.
Selected memories and shared references shape tasks. The Gomoku level reworks a prior game; the ring represents an imagined future rather than a past event.

D2 Staged access.
Ordered levels and door entry organise the narrative and pace normal play; they are not security barriers.

D3 Local progress.
Completion state is stored on-device and shared through a code, without an author-operated progress backend. External services remain involved in other features.

**Table 2: Design commitments and the mechanisms that realise them.**

| Commitment | Realised by |
|---|---|
| D1 Memory-derived mechanics | Memories, cultural references and imagined futures become tasks (Table 3) |
| D2 Earned access | Ordered level registry; router guard; door-only entry; cats guarding the next locks |
| D3 Local progress | `localStorage` per player; grow-only set; progress codes; PWA offline cache |
| D4 Bilingual coverage | Matching non-empty dictionary keys; no independent language-quality evaluation |
| D5 Drawn in code | Sprites and tiles as strings; banner and icons generated from them |
| D6 Delayed disclosure | Lexical test over eligible dictionary keys, with explicit exclusions |

D4 Bilingual coverage.
Both dictionaries contain non-empty values for the same keys.

D5 Code-defined pixel art.
Town tiles and character sprites are generated in code. Photographs, audio, rendered rings and other assets have separate provenance and reuse conditions.

D6 Delayed narrative disclosure.
A lexical test checks dictionary entries shown before the finale for explicit references to it.

# 4 The Experience

## 4.1 The Front Door

The site opens on a single question. The answer, a date both players know, is accepted in several spellings: its digits are compared against SHA-256 hashes [24]. Writing $\delta(a)$ for the digits of an answer $a$ in order, $H$ for SHA-256 and $\mathcal{A}$ for the four stored digests,

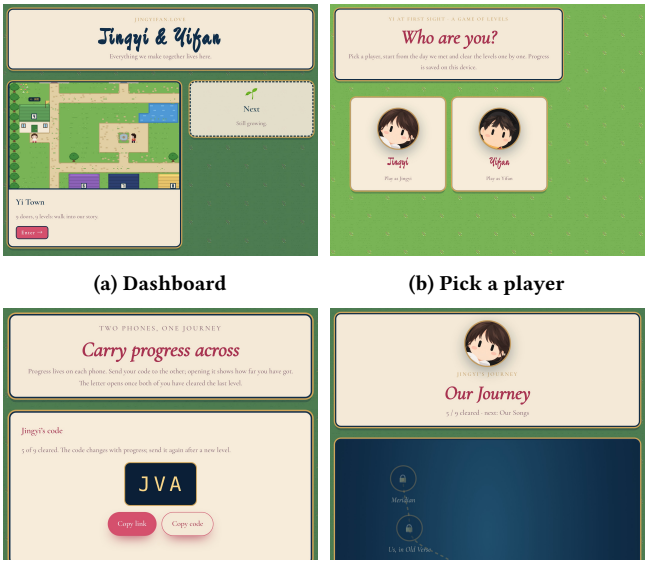
$$\text{pass}(a) \iff H\big(\delta(a)\big) \in \mathcal{A}. \tag{1}$$

Hashing changes how the answer is compared, but the date also appears in the game's own narrative; it is a recognition question, not a secret. Behind it is a dashboard that lists everything the couple build under the domain (Figure 2a), of which Yi Town is the first. Choosing the town asks *Who are you?* (Figure 2b). Progress is kept per player on each device, so both can play on one phone.

## 4.2 The Town

The town (Figure 1a) is a $24 \times 20$ grid of $16 \times 16$-pixel tiles. Paths connect nine houses around a fountain square and a pond. Each roof has its own colour and its level number, and a cleared house flies a gold flag. The player moves with the arrow keys, by tapping a tile (the avatar walks a shortest path to it), or with an on-screen d-pad. Walking into a door opens a talk box with the level's one-line story, for example "Our first game was on a Zoom whiteboard. Now you place the last stone."

Four kinds of character live in the town (Figure 3). The *partner* waits by the fountain and reports how far they have got ("I am waiting here. I have cleared 5 levels. You?"). *Rayford*, Yifan in a suit, is a scripted guide south of the fountain: he explains the project,




**(a) Dashboard** **(b) Pick a player**

**(c) Progress code** **(d) The journey list**

**Figure 2: Around the town. (a) The dashboard lists every product under the domain. (b) Each device keeps one progress per player. (c) Progress crosses phones as a code (here JVA). (d) The journey as a list, for players who prefer it to the map.**



**Figure 3: Every character in Yi Town is a string of characters in the source. From left: Jingyi, Yifan, Rayford, Yifan as the baker, Jingyi crowned, the bride and the groom (four facings each), then the cats Coffee and Duoduo. Bodies are $12 \times 12$ pixels over four rows of shared legs.**

lists the levels, hints at the next one and walks the player to its door. Two Ragdoll cats, *Coffee* (咖啡) and *Duoduo* (哆哆), guard the locks: Coffee sits at the next locked door and Duoduo at the one after. When a door opens they walk to their new posts, and with nothing left to guard they sunbathe by the pond. Bumping into a cat shows a photo of the real cat. Two signposts lead out of the square to the two cities (Section 4.4), and a red mailbox holds voice notes the partner has left.

The sky follows the clock in the player's own city: Jingyi walks the town in Shenzhen time and Yifan in College Station time, so at the same instant one sees noon and the other sees lit windows under a night wash. The whole site stands on the same pixel meadow, and every heading is a cream sign in the town's palette, so leaving a house never leaves the world.

## 4.3 Nine Levels

Table 3 lists the levels with their source memory, mechanic and input modality, and Figure 4 shows them in play.

*1. The Day We Met.* The first level shows the day count since 20 September, the time in both cities with day and night icons, and a countdown to the flight. To clear it, each player reads a vow, rewritten from the wedding formula as the vow of "becoming each other's partner once more", and says one line aloud: 我爱你，陈静仪，我愿意 ("I love you, Jingyi Chen, I do"), with the partner's full name. Browser speech recognition [34] hears Mandarin names inconsistently, so the matcher removes everything except Han characters and Latin letters and accepts any of eight or nine homophone spellings of each name (for example 陈静怡, 陈静宜). A typed fallback keeps the level playable in browsers that cannot listen.

*2. Love Map.* Six places matter to the couple: Texas, Los Angeles, Japan, Shenzhen, Hong Kong and New York. The street views are public photographs of well-known sites from Wikimedia Commons rather than the couple's own pictures, since a personal photograph can give away far more of a place than its owner intends [35]. Before each heart appears on a Leaflet map, the player names a street-level photograph from four options (Figure 4b), and a wrong pick reshuffles the options. Great-circle arcs connect the two home cities.

*3. Tattoo.* The player handwrites a line or a small drawing on an arm or a shoulder, drawn in SVG, and inks it. The tattoo is kept on the device and can be saved as an image to send.

*4. Moments.* The timeline holds eight dated memories, each with photos and a story in both languages, from hearts made from both sides of the ocean (day 21) to a heart drawn on the Gomoku board (day 35). The player orders five of them by tapping pairs to swap, then reassembles a $3 \times 3$ portrait of the partner.

*5. Gomoku.* Each of ten positions on a $15 \times 15$ board has exactly one winning point for the player's own colour (purple for Jingyi, green for Yifan, as in their first game). The ten answers are the ten points of a heart; solved answers stay on the board in gold, so the heart draws itself as the player progresses. Positions are generated from three patterns (a four closed at one end, a broken four, three-and-one) placed around each heart point in varied directions with decoy stones (Figure 5). Writing $M$ for the mover's stones, $\mathcal{E}$ for the empty cells and $\mathrm{five}(\cdot)$ for "contains five in a row", a test requires of every position $k$ that

$$\left|\{\, c \in \mathcal{E} : \mathrm{five}(M_k \cup \{c\})\,\}\right| = 1, \qquad \{c_1^*, \ldots, c_{10}^*\} = \mathcal{H}, \tag{2}$$

where $c_k^*$ is that unique cell and $\mathcal{H}$ the ten cells of the heart. Gomoku is a first-player win under optimal play [1]; win-in-one puzzles avoid that question entirely and make every move feel like a find.

*6. Our Songs.* Six songs the couple share (among them 只对你有感觉 and 天下第一好) play a passage through the official embedded players: QQ Music for the player in China and YouTube for the player in the United States, each switchable. The site hosts no audio. Embedded players expose no playback events, so the options unlock after eight seconds on screen, whether or not the passage was heard.

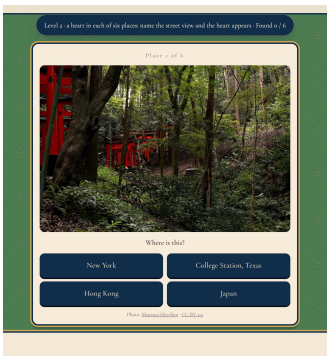
*7. Libra and Scorpius.* Yifan is a Libra (25 September) and Jingyi a Scorpio (31 October); in the sky the two constellations are neighbours, and Libra's two brightest stars still bear the Arabic names of the Scorpion's claws, *Zubenelgenubi* and *Zubeneschamali*. The level projects 21 manually entered stars, six for Libra and fifteen for Scorpius, and asks the player to draw 20 edges. The source records

**Table 3: The nine levels: autobiographical material, shared references and imagined futures become tasks (D1). Modalities: *V* voice, *T* touch or pointer, *C* camera, *K* keyboard alternative available.**

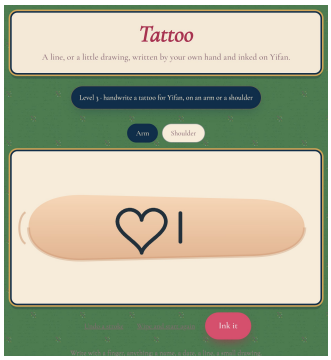
| # | Level | Design material | Task (clear condition) | Input | Data |
|---|---|---|---|---|---|
| 1 | The Day We Met | 20 September 2026; the vow | Answer a question, then say 我爱你，<name>，我愿意 aloud; homophones of the name accepted | V/K | 1 vow per player |
| 2 | Love Map | Where each studied; where they will meet | Name six street views; each right answer drops a heart on the map to collect | T | 6 places |
| 3 | Tattoo | A promise to write on each other | Handwrite a line on an arm or shoulder, then "ink it" | T | strokes |
| 4 | Moments | The first 40 days | Order five memories, then reassemble a 3 × 3 portrait of the partner | T | 8 memories |
| 5 | Gomoku | First game on a Zoom whiteboard | Ten win-in-one endgames; the ten answers form a heart | T | 10 positions |
| 6 | Our Songs | The six songs they share | Hear a passage, pick the title from four | T | 6 songs |
| 7 | Libra and Scorpius | Their birth signs, neighbours in the sky | Join 21 stars along 20 edges | T/K | J2000 positions |
| 8 | Us, in Old Verse | Classical love poetry | Restore the missing half-line of five poems character by character | T | 5 poems |
| 9 | Meridian | What comes next | Try the ring on through the camera and take a photo | C | 3 GLB models |



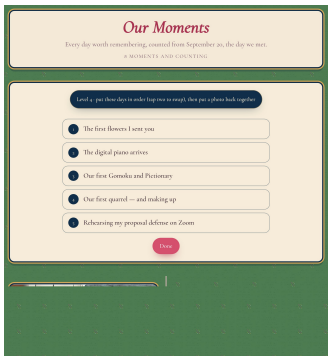

**(a) 1 The Day We Met**



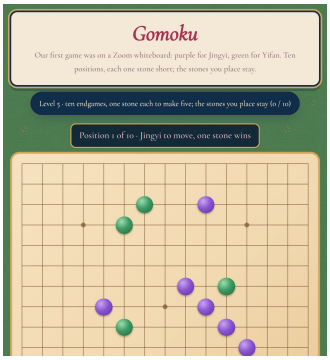

**(b) 2 Love Map**




**(c) 3 Tattoo**




**(d) 4 Moments**




**(e) 5 Gomoku**




**(f) 6 Our Songs**




**(g) 7 Libra and Scorpius**

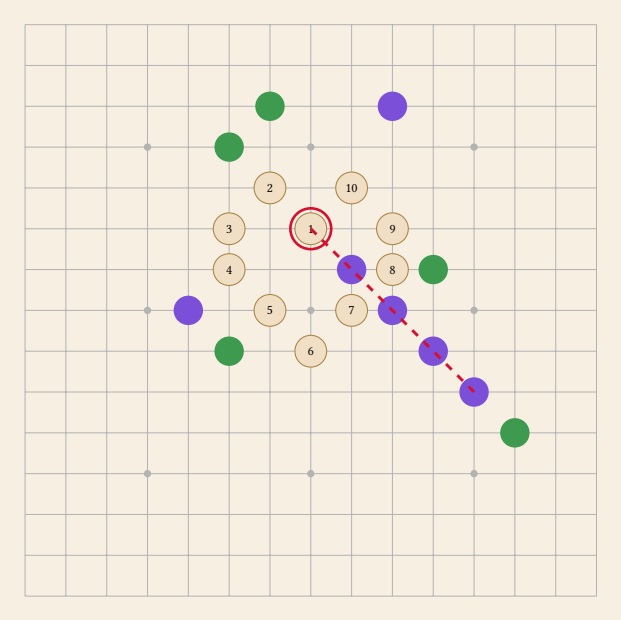



**(h) 8 Us, in Old Verse**

**Figure 4: Eight of the nine levels in play (the ninth is shown in Figure 6). (a) Day counter, twin clocks and the countdown to CYF925. (b) Name the place in the street view (photograph: Maarten Heerlien, CC BY 2.0, via Wikimedia Commons). (c) A heart inked on an arm. (d) Put the first days in order (the photographs below are omitted here). (e) Position 1 of 10: one stone wins. (f) Name the song. (g) Libra's "claws" and Scorpius, before they are joined. (h) A line traditionally attributed to Su Wu, 结发为夫妻，恩爱两不疑 with its second half to restore.**

coordinates as J2000 right ascension and declination, following the conventions in Meeus [23]; the chart is an illustration, not a view of either player's current night sky.

*8. Us, in Old Verse.* Five classical lines about love lose their second half, to be restored from a bank of characters: a farewell traditionally attributed to Su Wu, 结发为夫妻，恩爱两不疑; Guan Daosheng's *Song of You and Me*; a couplet of Li Shangyin; the paired birds of the




15 × 15 board, rows 4–8 of the heart highlighted; purple: mover (Jingyi), green: partner

**Figure 5: Ten answers draw a heart. The gold discs are the ten cells of $\mathcal{H}$ in the order they are found (Eq. 2); the stones are position 1 as generated in the source (`heart-1`): six purple stones for the mover, five green decoys, and the unique winning cell (red ring) completing a diagonal five. Later positions are built the same way around the other nine cells, so solved answers accumulate into the outline.**

sixth-century anthology *New Songs from a Jade Terrace* [4]; and an anonymous couplet. Each solved line shows a plain English gloss.

*9. Meridian.* The last level is a pair of rings, called *Meridian*, modelled procedurally in Blender from a Python script (Figure 6). The pair share a signature: a yellow-gold line inlaid around a platinum band like a meridian on a globe, a hidden sapphire inside the bottom, and an inner engraving that carries the *other* person's name and birthday, the date they met and the coordinates of the airport where they will meet (40.64°N 73.78°W). Her ring exists in two designs, a solitaire on four compass prongs (v1) and the current flush-set stone sunk into his band's profile (v2); the page keeps both as a design history. The level is cleared only one way: the player tries the ring on through the camera and takes a photo (Section 5.5).

## 4.4 Two Days

Two signposts lead from the square to a day in each city (Figure 7). Each is a second map with the person's stops: Yifan's home, gym, his advisor's office, the golf course and the pool in College Station; Jingyi's home with the cats at the door, her drive, the office and Shenzhen Bay. Each stop has hours, and the person's sprite stands wherever the clock in their own city puts them. A partner who opens Yifan's day at 9:49 p.m. in Shenzhen finds him at 8:49 a.m. at his advisor's office. The schedules are authored, not sensed; they began as guesses and were corrected by the designer: Yifan drives

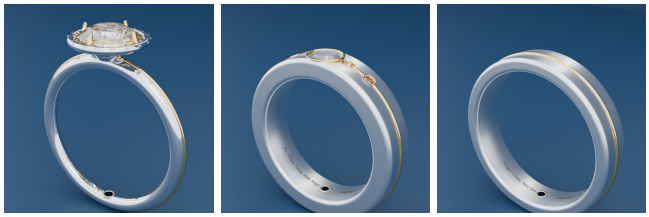


**Figure 6: The Meridian pair, path-traced from the procedural Blender model: hers v1 (solitaire, halo and graduated shoulder stones), hers v2 (flush-set 4.2 mm stone in a 5 mm flat band), and his (brushed flat comfort-fit band). The gold meridian runs around each band.**




**(a) A day of Yifan's** **(b) A day of Jingyi's**

**Figure 7: The two cities as maps. The person stands at the stop whose hours contain the current local time in their own city: (a) Yifan at his advisor's office at 8:49 a.m. in College Station; (b) Jingyi at home at 9:49 p.m. in Shenzhen, the same instant.**

rather than taking the bus; Jingyi drives a Tesla rather than taking the metro; and the Shenzhen Bay stop keeps a joke between them about where they will one day live.

## 4.5 Prologue and Finale

Two cutscenes frame the game (Figure 8). They run in a small pixel theatre: a 128 × 80 canvas scaled without smoothing, actors built from the town's sprite sheets, a darkness layer with candle-like glows, particles, a synthesised score, and a caption box for words and buttons.

*The birthday prologue* plays once per player before the town opens. In College Station it is 9:47 a.m.; in Shenzhen 10:47 p.m., 73 minutes before Jingyi's birthday. Yifan, in a chef's hat, bakes the cake against the Shenzhen clock. The player helps by whisking the batter (twelve taps), placing ten strawberries in the same heart that the Gomoku answers make, and watching nine candles lit (nine levels; in Chinese 九 "nine" sounds like 久 "for a long time"). At the prologue's scripted Shenzhen midnight the lights go out, the door opens on the town ("In Yi Town there is no thirteen-hour time difference"), everyone sings, the player makes a wish and blows the candles out, by breathing into the microphone or holding a button.

*The wedding* opens after the last letter, once both players have cleared the last level. In the fountain garden at dusk, Rayford reveals why he had worn a suit all along: he is the master of ceremonies. The groom waits under the arch, the bride walks the aisle to Mendelssohn's *Wedding March*, the scripted sequence asks for both characters' "I do" responses, with speech recognition where supported, Coffee carries the rings down the aisle on a red pillow, the Meridian rings turn in 3D, and the scene ends in a kiss, bells, doves and a certificate to keep (Figure 9). Afterwards the town dresses the two sprites as bride and groom.

# 5 System

## 5.1 Architecture

Yi Town is a static single-page application in strict TypeScript, built with Vite and served from static hosting under the couple's domain (Figure 10). The deployment configuration targets GitHub Pages. There is no author-operated backend for progress. Each feature is a folder (`src/features/<name>/`) with its page module, styles, data, tests and a README, and each page is a function `render(root)` $\Rightarrow$ `cleanup` loaded lazily by a path router, so the main bundle carries only the shell and the town. Heavy modules (Leaflet for the map, three.js for the ring, MediaPipe for the try-on) load only inside the level that needs them. A service worker precaches the shell and caches eligible media after retrieval. The cached town can run offline, but remote song embeds and browser-dependent speech recognition are not guaranteed offline; cache eviction and previously unfetched assets also limit availability.

## 5.2 The Quest Model

Let $L = (\ell_1, \ldots, \ell_9)$ be the ordered level registry and $C_p \subseteq L$ the set of levels player $p$ has cleared on a device. A level is unlocked iff it is first or its predecessor is cleared:

$$\text{unlocked}_p(\ell_i) \iff i = 1 \ \vee \ \ell_{i-1} \in C_p. \tag{3}$$

Clearing adds to $C_p$ and nothing ever removes from it, except an explicit reset. The next level to play is the first one not yet cleared, $\text{current}(p) = \ell_{\min\{i : \ell_i \notin C_p\}}$. The router enforces the lock, and a second guard requires that a level be entered through its door: walking onto the door sets a one-shot token $\tau$ in `sessionStorage`, and a level route without the token redirects to the town, so that

$$\text{enter}_p(\ell_i) \iff \text{unlocked}_p(\ell_i) \ \wedge \ \tau = \text{route}(\ell_i). \tag{4}$$

Figure 11 shows the whole sequence, including the two scenes gated on both players. Level numbers are derived from the registry, so inserting a level (the Poems level was inserted before the last one during development) renumbers every house, sign and string at once; a test fails until the new house has a door that the town's path search can reach.

## 5.3 Progress without a Server

Each phone holds its own $C_p$. Because $C_p$ only grows, it is a grow-only set (G-Set), the simplest state-based CRDT [29]: the merge of two replicas is their union,

$$C_p \sqcup C'_p = C_p \cup C'_p, \tag{5}$$

**Table 4: Example progress codes: player letter, base-36 mask (Eq. 6) and the check character of Eq. 7.**

| Cleared | Mask $m$ | Base 36 | Code |
|---|---|---|---|
| Jingyi, levels 1–5 | 31 | `V` | `JVA` |
| Jingyi, all nine | 511 | `E7` | `JE7Z` |
| Yifan, all nine | 511 | `E7` | `YE7O` |
| Yifan, none | 0 | `0` | `Y0C` |

which is commutative, associative and idempotent. Replicas converge if all updates are eventually exchanged, both phones use the same level registry and no reset occurs. Repeated imports are idempotent; an old code cannot remove completion flags, but incomplete exchange can leave the two views different. The only channel needed is one the couple already use every day: a chat message.

*Progress codes.* A code encodes one player and a bitmask of cleared levels, written in base 36 after a player letter $\lambda_p \in \{\texttt{J}, \texttt{Y}\}$ and followed by one check character $\chi$ (Figure 12):

$$m = \sum_{\ell_i \in C_p} 2^{\,i-1}, \qquad \text{code}(p, C_p) = \lambda_p \,\|\, \beta_{36}(m) \,\|\, \chi\big(\lambda_p \| \beta_{36}(m)\big), \tag{6}$$

where $\|$ is concatenation and $\beta_{36}$ writes a number in base 36. The decoder accepts a string only if its last character equals $\chi$ of the rest, its first is a player letter and $0 \le m < 2^9$. With nine levels $m \le 511 = \text{E7}_{36}$, so a code is at most four characters: `JVA` says that Jingyi has cleared $m = 31$, the first five levels (Table 4). The check character follows the hybrid system ISO 7064 MOD 37,36 [13]. Writing $a_j \in \{0, \ldots, 35\}$ for the value of the $j$-th body character and starting from $p_0 = 36$,

$$s_j = \big((p_{j-1} + a_j) \bmod 36\big) \text{ or } 36 \text{ if } 0, \qquad p_j = 2s_j \bmod 37, \tag{7}$$

and the check character is the symbol of value $(37 - p_n) \bmod 36$. We checked the scheme exhaustively over all 1,024 valid codes: 140,840 single-character substitutions and 2,943 transpositions of two different adjacent characters. It rejects every substitution. Of the transpositions it rejects all but five. One of those (`JI0`$\rightarrow$`IJ0`) is still refused because `I` is not a player letter. The other four all swap a `0` and a `1` inside the mask (`J801`$\leftrightarrow$`J810`, `YE01`$\leftrightarrow$`YE10`): the rule above maps a running sum of 0 to 36, so a zero digit and its neighbour can trade places unseen. An accepted typo may add unearned completion flags and unlock later content, including the finale. An earlier version used a plain weighted sum, which let some one-character typos through; it was later replaced, invalidating older codes. Codes are not authenticated: a person can construct completion for either player. They carry no schema version, so registry insertions or reordering can reinterpret bits. Reset is outside the grow-only model; importing an earlier code after reset restores its flags. A code may also travel in a link (`/sync?c=JVA`), exposing its payload to the hosting request and potentially to browser history or the channel through which it is sent.

*Three layers of storage.* Cleared levels live in `localStorage` under a key per player; half-finished state inside a level (the Gomoku position reached, the stars joined so far) lives in `sessionStorage`, so a reload or a language switch inside a level does not restart it, and it




**Figure 8: The two cutscenes, left to right. Top, the birthday prologue: the kitchen at 9:47 a.m. in College Station (10:47 p.m. in Shenzhen) with both clocks on the wall; whisking the batter; nine candles on the finished cake; singing at Shenzhen midnight with the cats. Bottom, the wedding: the guests take their seats; the bride walks the aisle; Coffee brings the rings; the Meridian rings in 3D over the ceremony.**




**Figure 9: The certificate at the end of the wedding, saved as an image through the phone's share sheet. It records the day of the ceremony, the day they met and the day they will meet.**

is dropped when the level is cleared; and a service worker keeps the site itself. When the browser blocks storage, an in-memory store keeps progress for the visit.

## 5.4 The Town Engine

The town is a grid of characters (. grass, # path, ~ water, T tree, 0 fountain, and so on). Movement is one tile per 190 ms with an ease-out interpolation $e(t) = 1 - (1-t)^2$ of the step's progress $t \in [0, 1]$; the pure functions `step` and `faced` decide movement and facing and are unit-tested, while a class wires them to a canvas and one animation loop. A tap on the map runs a breadth-first search over walkable tiles and queues the resulting steps. The camera follows the player. The same engine takes any map, so the two cities reuse it with their own grids, roads and a basketball court.

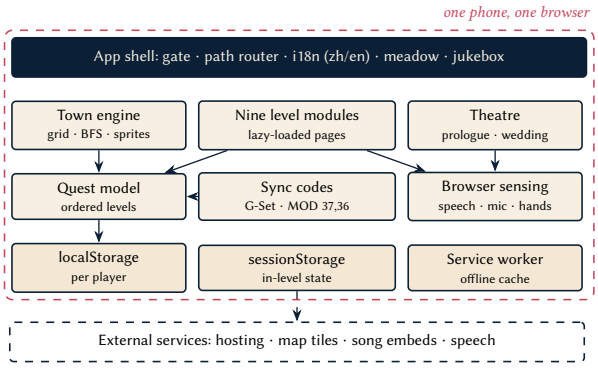



**Figure 10: Architecture of application modules and local state. The dashed boundary marks browser components, not a confidentiality boundary: static assets are hosted remotely, external requests expose request metadata, and speech recognition may use a remote service. Progress crosses devices through a manually exchanged code.**

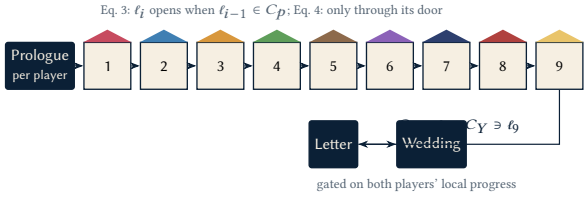



**Figure 11: The sequence of play. Nine houses open in order per player; the letter and the wedding open only when both players' completion sets on the device contain the last level.**

*Art as text (D5).* A sprite is an array of strings, one character per pixel, and a palette that maps characters to colours. A figure is a 12 × 12 body per facing over four rows of legs shared between

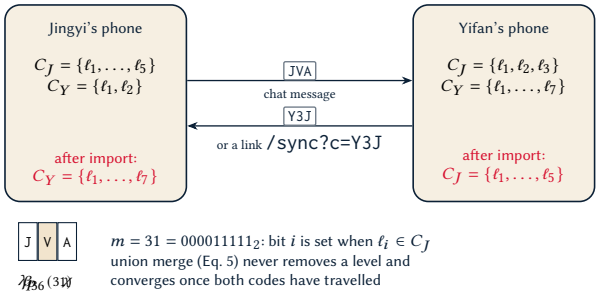



**Figure 12: Progress without a server. Each phone keeps both players' completion sets; a short code (Eq. 6) carries one set across in a chat message or a link, and the receiver takes the union (Eq. 5). Below: the three fields of `JVA`.**

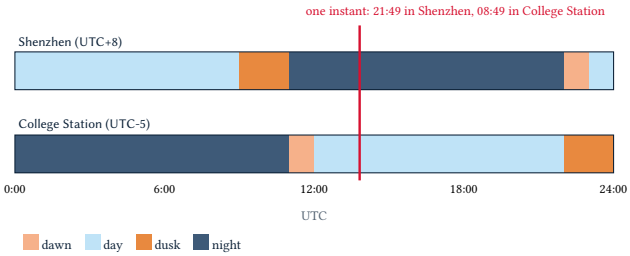



**Figure 13: The two skies over one UTC day (Eq. 8), with the thirteen-hour offset in force during development. Night in one town overlaps day in the other for most of every day; the red line is the instant shown in Figure 7.**

facings, so a walk cycle is the same two leg frames under different bodies, and the side view is mirrored. Tiles (grass with flowers, paths that join their neighbours, roofs in each level's colour, lit windows) are painted procedurally. The README banner, the favicon and the app icons are generated from the same strings by scripts, so a new outfit appears everywhere at once.

*Daylight by city.* Let $h_p$ be the current hour in player $p$'s time zone (`Asia/Shanghai` or `America/Chicago`). The phase of the sky and the person's place in a city day are

$$\phi(h) = \begin{cases} \text{dawn} & 6 \le h < 7, \\ \text{day} & 7 \le h < 17, \\ \text{dusk} & 17 \le h < 19, \\ \text{night} & \text{otherwise}, \end{cases} \tag{8}$$

$$\text{stop}(h) = \min\left\{\, j : \exists\, [a, b) \in W_j,\ a \le h \bmod 24 < b \,\right\}, \tag{9}$$

where $W_j$ is the set of hour windows of the $j$-th stop. Dawn, dusk and night each lay a coloured wash over the map (night: navy at 42% opacity) and light the windows. Because $h_J$ and $h_Y$ differ by the two cities' offset, the same instant falls in different phases for the two players (Figure 13).

## 5.5 Sensing in the Browser

Camera tracking and breath-amplitude processing run in application code on the device. Browser speech recognition has a different processing boundary.

*Voice.* The vow and the wedding's "I do" use the Web Speech API [34] in Mandarin. The API allows a remote recognition service unless local processing is explicitly required [34], and the implementation does not require it, so voice samples may leave the device. A typed fallback is available for the vow. Recognition alternatives are joined into one string $w$ and normalised by $\nu$, which keeps only Han characters and Latin letters; with $\sqsubseteq$ for "is a substring of" and $\mathcal{N}_p$ the accepted spellings of the partner's name,

$$\text{vow}_p(w) \iff \text{我爱你} \sqsubseteq \nu(w) \ \wedge\ \text{我愿意} \sqsubseteq \nu(w) \ \wedge\ \exists\, n \in \mathcal{N}_p :\ n \sqsubseteq \nu(w). \tag{10}$$

*Breath.* Blowing out the candles reads the microphone with echo cancellation, noise suppression and gain control off. For each frame $t$ of $N = 1{,}024$ samples $x_{t,i}$,

$$r_t = \sqrt{\tfrac{1}{N}\textstyle\sum_{i=1}^{N} x_{t,i}^2}, \qquad s_t = 0.6\, s_{t-1} + 0.4 \min(1, 5 r_t), \tag{11}$$

$$\text{blow}_t \iff s_t > 0.32.$$

We rely on a breath being a loud, broadband rush on a phone microphone; the threshold was set by hand and has not yet been calibrated across devices.

*Hands.* The try-on runs MediaPipe's hand landmarker [22, 36] on the camera stream, which returns 21 landmarks $\mathbf{k}_0, \ldots, \mathbf{k}_{20}$ per hand (Figure 14). The ring's axis $\mathbf{a}$ follows the ring finger's proximal phalanx, its "up" $\mathbf{u}$ is the back of the hand, taken from the palm normal $\mathbf{n}$ with the sign given by the reported handedness, and its centre $\mathbf{c}$ sits a fraction $\beta = 0.38$ along the phalanx:

$$\mathbf{a} = \frac{\mathbf{k}_{14} - \mathbf{k}_{13}}{\|\mathbf{k}_{14} - \mathbf{k}_{13}\|}, \qquad \mathbf{n} = \pm\, \frac{(\mathbf{k}_5 - \mathbf{k}_0) \times (\mathbf{k}_{17} - \mathbf{k}_0)}{\|(\mathbf{k}_5 - \mathbf{k}_0) \times (\mathbf{k}_{17} - \mathbf{k}_0)\|}, \tag{12}$$

$$\mathbf{u} = \frac{\mathbf{n} - (\mathbf{n} \cdot \mathbf{a})\, \mathbf{a}}{\|\mathbf{n} - (\mathbf{n} \cdot \mathbf{a})\, \mathbf{a}\|}, \qquad \mathbf{c} = (1 - \beta)\, \mathbf{k}_{13} + \beta\, \mathbf{k}_{14}, \tag{13}$$

and the rotation is the orthonormal basis $R = [\, \mathbf{u} \times \mathbf{a},\ \mathbf{u},\ \mathbf{a}\, ]$. Scale comes from anatomy: the mean spacing of the middle, ring and little knuckles is the ruler, set to a constant of 19.5 mm (a working estimate for an adult hand),

$$\text{px/mm} = \tfrac{1}{2}\left(\|\mathbf{k}_9 - \mathbf{k}_{13}\| + \|\mathbf{k}_{13} - \mathbf{k}_{17}\|\right) / 19.5, \tag{14}$$

so nominal model dimensions map to the screen at an assumed scale; the overlay is stylised and says nothing about ring size or fit. Position, rotation and scale are smoothed exponentially with $\alpha = 0.45$ per frame, $\mathbf{c}_t = \mathbf{c}_{t-1} + \alpha(\mathbf{c} - \mathbf{c}_{t-1})$ and $q_t = \text{slerp}(q_{t-1}, q, \alpha)$ for the rotation quaternion.

The shutter composes the photo with a caption, and a second card puts both players' try-on photos side by side.

## 5.6 Keeping a Secret with a Test (D6)

The ring is the surprise at the end, and an interface that hints at it spoils it. Rayford, the town, the progress page and the cats all speak before the ring is reached, and their lines were written and rewritten over days. Rather than rely on vigilance, a unit test scans every key in the bilingual dictionary, except the keys shown only after the ring, and fails if any contains 戒 (ring) in Chinese, *ring* as

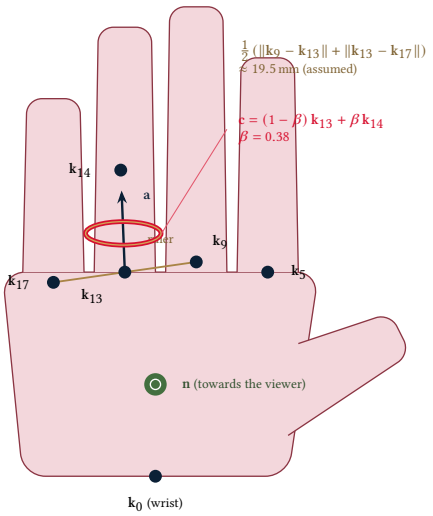



**Figure 14: Placing the ring from hand landmarks (Eqs. 12–14). The band's axis follows $\mathbf{k}_{13} \to \mathbf{k}_{14}$, its centre sits 38% along that segment, the back of the hand is the palm normal $\mathbf{n}$ oriented by handedness, and the knuckle spacing $\mathbf{k}_9$–$\mathbf{k}_{13}$–$\mathbf{k}_{17}$ is the millimetre ruler.**

a word in English, or any word for wedding, marriage, proposal, bride or groom in either language. With $K$ the set of dictionary keys, $K_{\text{after}}$ those shown only after the ring, $A$ the allow-list and $R_{\text{zh}}, R_{\text{en}}$ the two pattern tests, the test asserts

$$\forall\, k \in K \setminus (K_{\text{after}} \cup A): \quad \neg R_{\text{zh}}\big(D_{\text{zh}}[k]\big) \;\wedge\; \neg R_{\text{en}}\big(D_{\text{en}}[k]\big), \qquad (15)$$

and a companion test asserts coverage, $\forall k \in K : D_{\text{zh}}[k] \neq \varepsilon \wedge D_{\text{en}}[k] \neq \varepsilon$ (D4). One element of $A$ is listed by name: Jingyi's own joke on the Shenzhen Bay stop, which is about a flat, not a ring. The test guards the dictionary only; assets, routes, source code and visual cues lie outside it.

## 5.7 Implementation and Verification

Table 5 describes the version 1.17.0 implementation. An audit reran the existing suite: all 153 tests in 36 files passed, and the strict TypeScript check and production build succeeded. The main JavaScript chunk was 114.09 kB (42.58 kB gzip), and the service worker precached 44 entries totalling 302.12 KiB; these are build sizes, not load times on phones. The large ring, map and try-on chunks are lazy loaded.

The audit also independently enumerated the progress-code cases reported above. Dictionary tests check matching non-empty keys; puzzle tests check properties such as unique winning moves.

The development records describe browser checks at desktop, 390-pixel and 320-pixel widths and a laptop-camera check of hand orientation; no per-release end-to-end report is archived. Systematic phone checks of speech, breath, camera completion and sharing remain outstanding.

**Table 5: Implementation summary (version 1.17.0, 28 October 2026).**

| Measure | Value |
|---|---|
| TypeScript source, excluding tests (lines) | 8,414 |
| CSS (lines) | 1,080 |
| Bilingual dictionary (lines) | 791 |
| Test files / unit tests | 36 / 153 |
| Commits / versions (24–28 Oct. 2026) | 109 / 28 |
| Main bundle (min. / gzip) | 114 kB / 43 kB |
| Precached shell (service worker) | 44 files, 302 KiB |
| Feature folders | 24 |
| Runtime dependencies | 3 (Leaflet, three.js, MediaPipe) |
| Author-operated progress backend / app accounts | 0 / 0 |

# 6 Reconstructing Design Decisions

Four documented episodes make the rationale inspectable. Commit identifiers mark the recorded changes; the interpretations are ours.

*Replacing Pictionary (*`7d2c659`*, 26 October).* The implementation removes a same-device drawing game and adds a song-identification level. Our interpretation is that separate recognition of a shared song better accommodates different visit times than taking turns on one screen. A plainer reading is that the replacement reduced implementation complexity.

*Requiring a try-on photo (*`2bc3d07`*, 28 October).* The ring level's completion condition changes from viewing the reveal to capturing a try-on photo. This creates an artefact that can be shared, but also makes camera access and a capable device conditions for completing the narrative. The same mechanism can be read as participation or as an unwanted demand; only the recipient can say which.

*Correcting represented routines (*`1960404`*, 26 October).* The city-day descriptions change the transport and destination details to match the designer's account of the couple's routines. The record shows that plausible generated defaults required correction; whether the corrected version matches the other partner's own account is for the represented person to say.

*Revising the entry gate (*`ccabd31`*, 28 October).* This commit removes answer-revealing placeholder text and revises associated checks. The recognition question becomes less self-defeating; it remains a low-entropy, client-side date gate rather than access control.

Across these episodes, the recurring issue is how a technical rule expresses an assumption about the other person.

# 7 Discussion

## 7.1 From Representation to Action

The design links a shared reference to an action, then to a visible trace: a prior game becomes a win-in-one puzzle and its solution becomes one point of a heart. Compared with displaying the same memory as a caption, this adds an authored sequence of participation. The design question it raises is which part of a memory

can become a rule without losing its meaning. The case is one implementation of that move; whether it serves a couple better than a photo album, a message or another intimate artefact [11, 18] is open. The ring also exposes a boundary: an imagined future is not a remembered event, although both can organise narrative play.

### 7.2 Privacy Boundaries and Personal Material

The gate hashes a low-entropy date, whose value may also be inferred from the surrounding story. It deters casual entry but provides neither encryption nor dependable access control. Local completion flags avoid an author-operated progress database; they do not make hosted photographs, voice notes or letters private. The voice-note implementation, for example, references a static audio asset. Anyone authorised to obtain the deployed files or repository may obtain such material.

The browser boundary also differs by feature. Hand tracking and breath-amplitude calculation run locally, whereas speech recognition may contact a remote service. Tile and media requests expose request metadata to their providers. A local-first aspiration therefore needs a description of data flows, not only the absence of a custom backend [19]. A research release would ideally separate shareable code and synthetic examples from personal assets, since intimate images can also reveal location context [35]; the current repository does not make that separation.

### 7.3 Asynchrony, Temporal Representation and Staleness

Each player's sky uses the clock in their configured city, while the partner avatar represents locally stored progress. The design supports visiting at different times, but an avatar's state may be old until the next manual exchange. Its city-day position follows an authored schedule, not sensed presence. The distinction matters because a representation intended as companionship can otherwise imply knowledge of the partner's current activity that the system does not have. The birthday scene's scripted clocks and the thirteen- or fourteen-hour offset are temporal design materials [10, 27].

### 7.4 Staged Access, Surprise and Freedom to Decline

Ordered access and delayed disclosure coordinate a surprise, but also let one partner determine what the other must do before reaching a letter or finale. Requiring a photograph illustrates the cost: people without camera access, or who prefer not to provide an image, cannot clear that level through the current completion rule. Similarly, a scripted "I do" is a game response, not evidence of consent to a relationship commitment. The present sequence does not model refusal as a separate successful outcome. A future revision should distinguish declining an action from failing a task and should offer an equivalent path that preserves narrative access.

The lexical test is useful as a writing regression check, but it cannot ensure surprise in a client-delivered game. Source, routes, media, screenshots and this paper disclose the finale. Any study of surprise must therefore record whether the recipient encountered these materials before play. Ambiguity of this kind can itself shape anticipation [8].

### 7.5 Bilingual Coverage and Situated Meaning

The same-key test prevents missing dictionary entries, but does not establish equivalent meaning, tone or task difficulty. The Chinese title 仪见钟情 substitutes a character from Jingyi's name into "love at first sight"; the English version explains the reference. Likewise, classical Chinese character-restoration puzzles retain different language demands even when their prompts have English versions. What the test establishes is coverage; semantic quality and accessibility need a separate review, including by the intended recipient.

### 7.6 AI Assistance and Authorial Responsibility

The source and log document an AI-assisted implementation process in which the designer supplied requirements and corrected defaults. Each recorded correction marks a place where the designer disagreed with generated output; it says nothing about details that went uncorrected. The corpus lacks an unassisted comparison and a complete record of human and model effort, so it supports no productivity claim about five days of work. Personal representation, verification and publication decisions remain the authors' responsibility.

### 7.7 Limitations and Next Evidence

The principal limitation is evidential: developer playtesting is the only reported use, and no recipient experience, sustained practice or relationship outcome has been collected. The corpus is brief and selectively interpreted, with no independent analysis. Engineering checks are also bounded: tests do not establish deployment-wide privacy, all-browser compatibility or equivalent experience across languages; approximate hand scaling is not a fit measurement. Union merge depends on compatible registries and does not propagate resets. Remote media limits offline use and availability across regions.

A next stage could begin with a voluntary recipient walkthrough, recording what was understood, skipped, refused or reinterpreted, followed by a diary of revisits and code exchanges over time. If participants are recruited or data are collected under an institution, the applicable ethics determination must precede that collection. A larger sample alone would not repair missing accounts of situated meaning, authorial power or failure.

## 8 Conclusion

MERIDIAN documents the pre-deployment design of a two-person game that turns autobiographical material into tasks, places and staged ceremonies. Its development record makes four design revisions inspectable, and its implementation audit establishes selected software properties. These support a situated account of how memories can shape playable rules, not a claim that the game improves intimacy or that the recipient accepts its authored sequence. The central open question is how a personal gift can invite shared meaning while leaving the other person free to decline, reinterpret and reshape it.

## Ethics and Publication Scope

This manuscript reports design and implementation; it includes no interviews and no relationship evaluation, so no institutional

ethics review was sought. The two people whose names, dates, photographs, voices and daily routines appear in the game and in this paper are its two authors, and each has agreed to that disclosure. Any future study with participants would need an ethics determination before data collection.

## Code and Data Availability

The source is public at https://github.com/rayford295/yi-town (since 30 October 2026) and the game runs at https://jingyifan.love. The version inspected in this paper is 1.17.0, commit `55440ae`; development continues on the main branch. The code is released under the MIT License. The photographs, voice notes, letters and other personal material in the repository remain the authors' own: they are published so that the game can be read and run, not as a research dataset, and may not be redistributed. A recorded walk through the whole game is at https://youtu.be/C0-oPjvNzZk.

## Declaration of Generative AI Use

Claude (Anthropic), through Claude Code, wrote much of the software, drafted text and prepared figures; Codex (OpenAI) was used to inspect the source and check the text. The authors reviewed everything that appears here and are responsible for it.

## Acknowledgments

To Coffee and Duoduo, who guard the doors. The street-view photographs on the Love Map are from Wikimedia Commons and credited in the game; the *Wedding March* is a public-domain recording from the European Archive.

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